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### Q1. Program based on Classes and Objects - Initializaing Data member using Objects directly

**Score: 10.00** **TC: 3/3 passed**

**CODE**

```
import java.util.Scanner;

public class Student {

    int rollNo;
    String name;
    double marks;

    public void displayDetails() {
        System.out.println("Roll No: " + rollNo);
        System.out.println("Name: " + name);
        System.out.println("Marks: " + marks);
    }

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        Student student = new Student();

        student.rollNo = sc.nextInt();
        sc.nextLine();
        student.name = sc.nextLine();
        student.marks = sc.nextDouble();

        student.displayDetails();
    }
}
```

**INPUT**

```
101
Rahul
89.5
```

**OUTPUT**

```
Roll No: 101
Name: Rahul
Marks: 89.5
```

## Q2. Classes and Objects-Initialize Object Data Using Setter Methods

**Score: 10.00** **TC: 3/3 passed**

**CODE**

```
import java.util.Scanner;

public class Student {

    private int rollNo;
    private String name;
    private double marks;

    public void setRollNo(int rollNo) {
        this.rollNo = rollNo;
    }

    public void setName(String name) {
        this.name = name;
    }

    public void setMarks(double marks) {
        this.marks = marks;
    }

    public void displayDetails() {
        System.out.println("Roll No: " + rollNo);
        System.out.println("Name: " + name);
        System.out.println("Marks: " + marks);
    }

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        Student s = new Student();

        int rollNo = sc.nextInt();
        sc.nextLine();
        String name = sc.nextLine();
        double marks = sc.nextDouble();

        s.setRollNo(rollNo);
        s.setName(name);
        s.setMarks(marks);

        s.displayDetails();

        sc.close();
    }
}
```

**INPUT**

```
121
bala
96.4
```

**OUTPUT**

```
Roll No: 121
Name: bala
Marks: 96.4
```

### Q3. Student Details Using Default and Parameterized Constructors

**Score: 10.00** **TC: 3/3 passed**

**CODE**

```
import java.util.Scanner;

public class Student {

    int rollNo;
    String name;
    double marks;


    Student() {
        rollNo = 101;
        name = "Rahul";
        marks = 85.5;
    }


    Student(int rollNo, String name, double marks) {
        this.rollNo = rollNo;
        this.name = name;
        this.marks = marks;
    }

    void displayDetails() {
        System.out.println("Roll No: " + rollNo);
        System.out.println("Name: " + name);
        System.out.println("Marks: " + marks);
    }

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);


        Student s1 = new Student();
        s1.displayDetails();


        int rollNo = sc.nextInt();
        sc.nextLine();
        String name = sc.nextLine();
        double marks = sc.nextDouble();


        Student s2 = new Student(rollNo, name, marks);
        s2.displayDetails();

        sc.close();
    }
}
```

**INPUT**

```
102
Ravi
90.5
```

**OUTPUT**

```
Roll No: 101
Name: Rahul
Marks: 85.5
Roll No: 102
Name: Ravi
Marks: 90.5
```

### Q4. Net Salary Calculation Using Single Inheritance

**Score: 10.00** **TC: 3/3 passed**

**CODE**

```
import java.util.Scanner;

public class Employee {

    int empId;
    String empName;
    double basicSalary;

    void getEmployeeDetails(int empId, String empName, double basicSalary) {
        this.empId = empId;
        this.empName = empName;
        this.basicSalary = basicSalary;
    }

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int empId = sc.nextInt();
        sc.nextLine();
        String empName = sc.nextLine();
        double basicSalary = sc.nextDouble();

        Salary s = new Salary();

        s.getEmployeeDetails(empId, empName, basicSalary);
        s.calculateSalary();
        s.displayDetails();

        sc.close();
    }
}

class Salary extends Employee {

    double hra;
    double da;
    double netSalary;

    void calculateSalary() {
        hra = basicSalary * 0.20;
        da = basicSalary * 0.10;
        netSalary = basicSalary + hra + da;
    }

    void displayDetails() {
        System.out.println("Employee ID: " + empId);
        System.out.println("Employee Name: " + empName);
        System.out.println("Basic Salary: " + basicSalary);
        System.out.println("HRA: " + hra);
        System.out.println("DA: " + da);
        System.out.println("Net Salary: " + netSalary);
    }
}
```

**INPUT**

```
192
bala
100000
```

**OUTPUT**

```
Employee ID: 192
Employee Name: bala
```

```
Basic Salary: 100000.0
HRA: 20000.0
DA: 10000.0
Net Salary: 130000.0
```

### Q5. Employee Gross Salary Calculation Using Multilevel Inheritance

**Score: 10.00** **TC: 3/3 passed**

**CODE**

```
import java.util.Scanner;

public class Employee {

    int empId;
    String empName;

    void getEmployeeDetails(int empId, String empName) {
        this.empId = empId;
        this.empName = empName;
    }

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int empId = sc.nextInt();
        sc.nextLine();
        String empName = sc.nextLine();
        double basicSalary = sc.nextDouble();

        GrossSalary gs = new GrossSalary();

        gs.getEmployeeDetails(empId, empName);
        gs.getBasicSalary(basicSalary);
        gs.calculateSalary();
        gs.displayDetails();

        sc.close();
    }
}

class Salary extends Employee {

    double basicSalary;

    void getBasicSalary(double basicSalary) {
        this.basicSalary = basicSalary;
    }
}

class GrossSalary extends Salary {

    double hra;
    double da;
    double grossSalary;

    void calculateSalary() {
        hra = basicSalary * 0.20;
        da = basicSalary * 0.10;
        grossSalary = basicSalary + hra + da;
    }

    void displayDetails() {
        System.out.println("Employee ID: " + empId);
        System.out.println("Employee Name: " + empName);
        System.out.println("Basic Salary: " + basicSalary);
        System.out.println("HRA: " + hra);
        System.out.println("DA: " + da);
        System.out.println("Gross Salary: " + grossSalary);
```

```
    }
}
```

**INPUT**

```
101
Rahul
30000
```

**OUTPUT**

```
Employee ID: 101
Employee Name: Rahul
Basic Salary: 30000.0
HRA: 6000.0
DA: 3000.0
Gross Salary: 39000.0
```